Pulidindi Maneesh Bhaskar

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LinkedIn | GitHub | LeetCode | Kaggle | CodeChef | Codeforces

Education

Indian Institute of Technology (IIT), Bhilai **2022 – 2026**
B.Tech in Computer Science & Engineering **CGPA: 8.27 / 10**

Bhashyam Junior College, Guntur **2020 – 2022**
Class XII (Intermediate) **77.5%**

Industrial Experience

Research AI Engineer — Frinks.ai (Industrial Machine Vision) May 2025 – Present

Converted from intern to full-time; embedded across Research and Deployment teams.

- **Scene Text Detection & Reading Direction (Research):** Researched and developed a hybrid **PSENet + TextSnake** architecture to achieve robust, reading-direction-aware text detection, replacing a failing legacy pipeline. Explored combining **RF-DETR** with a Bezier head for curved text, and conducted deep comparative studies of **CCD, SIGA, DPText-DETR, DeepSolo, ABCNet v1/v2** for direction-aware scene text recognition. Used **CRAFT** for annotation and ground-truth generation in research experiments.
- **End-to-End OCR Pipelines (Production):** Built and maintained production OCR detection pipelines running under **40ms** latency, deployed on **Control Print** and related industrial quality-control use cases. Worked on training and fine-tuning **PP-OCRv5** for multi-lingual recognition tasks.
- **Anomaly Detection — Sandpaper & Custom Surfaces:** Designed a **DINOv2-based RAG** anomaly detection system using patch-level embeddings and prototype retrieval to detect surface defects on sandpaper and other custom industrial materials, without requiring large labelled defect datasets.
- **Dimensions Measurement System (Production):** Contributed to a precision object dimensioning pipeline using **U^2-Net** for background matting and camera calibration, achieving sub-**100 microsecond** measurement latency for industrial throughput requirements.
- **Data Recommender for Training:** Built a data recommendation system to intelligently curate and prioritize training samples, improving model training efficiency across computer vision pipelines.
- **Deployment:** Worked on model deployment and inference optimization (ONNX, TensorRT) across production pipelines, ensuring real-time performance constraints are met on edge and cloud infrastructure.
- **Technologies:** PyTorch, TensorFlow, OpenCV, YOLO, PSENet, TextSnake, RF-DETR, DPText-DETR, DeepSolo, ABCNet, CCD, SIGA, CRAFT, DINOv2, U^2-Net, PP-OCRv5, ONNX, TensorRT, AWS, Python

Web Development Intern — TeKreant May 2023 – Jun 2023

- Contributed to full-stack development of a financial management web application.
- Built user-friendly interfaces with HTML, CSS, JavaScript, React.js, and Bootstrap; assisted in server-side development using Node.js, Express.js, and MongoDB/MySQL.
- Integrated RESTful APIs, implemented authentication/authorization features, and optimized website performance.

Research Experience

Research Intern — Indian Army (Inventory Management System) Apr 2024 – Jul 2025

- Designed and implemented a secure, real-time Inventory Management System for the Indian Army, enabling accurate tracking of critical materials and reducing manual overhead.
- Delivered a scalable, cost-efficient solution deployable across multiple depots and diverse operational environments with minimal hardware modifications.

Research Intern — Earthnet Resources Pvt. Ltd. (Mentored by Dr. R. Jose Immanuel, IIT Bhilai) Sep 2024 – Feb 2025

- Developed a real-time computer vision system for automated material separation, enabling classification by size, shape, color, and material type.
- Integrated SegNeXt-B2 CNN with LSTM to exploit spatial-temporal patterns, achieving notable improvements in classification and sorting accuracy.
- Engineered a scalable, cost-effective solution adaptable to diverse materials with minimal hardware reconfiguration, supporting deployment in varied industrial environments.
- **Technologies:** NVIDIA Jetson, Intel RealSense, PyTorch, YOLO, OpenCV, Open3D, TensorFlow, Keras, Raspberry Pi, Arduino, MATLAB, AWS, SAM

Research Assistant — IIT Bhilai (Under Prof. Dr. Vishwesh Jatala) 2023 – 2024

- Proposed an efficient Graph Partitioning and Compression-Based GNN Training Framework for large-scale web graphs.
- Developed a GPU-based multi-level graph partitioning system, compressing background nodes to accelerate target-node classification.
- Implemented GPU-accelerated coarsening, partitioning, and refinement phases, significantly outperforming METIS.
- Designed graph coarsening algorithms preserving spectral properties, ensuring minimal information loss while reducing GNN training and prediction times.
- **Technologies:** CUDA, Graph Partitioning, GNNs, Spectral Analysis, METIS, GPU Programming

Research Assistant — IIT Bhilai (Under Prof. Dr. Soumajit Pramanik) Apr 2024 – Jul 2024

- Developed **DeepSuccess**: a Seq2Seq deep learning framework predicting future success of Meetup groups using group size, event topics, and popularity trends.
- Designed a GNN-based baseline model using historical user-event interaction graphs, leveraging heterogeneous graph structures and affinity score computation.
- Achieved **87% accuracy**, outperforming state-of-the-art baselines by **8%**.
- **Technologies:** Deep Neural Networks, GNNs, Seq2Seq Models, Heterogeneous Graphs, Social Network Analysis, PyTorch

Projects

- **Depth-Sensing Based Volume Estimation** | *YOLOv5, Intel RealSense L515, Open3D, OpenCV, Python* GitHub
  - Built a real-time volume estimation system using YOLOv5 and Intel RealSense, achieving **89% accuracy** and reducing manual measurement time by **60%** across 200+ objects.
  - Reconstructed 3D point clouds using Open3D; optimized segmentation workflow for **30% faster frame processing** and 12% better boundary accuracy.
  - Applicable in logistics, warehouse inventory management, and agri-tech yield estimation.
- **EEG Classification and Image Generation** | *PyTorch, TensorFlow, Keras, CNN, LSTM, SNGAN* GitHub
  - Classified human EEG signals using CNN-LSTM hybrid models, achieving **85% accuracy** for early neurological pattern detection.
  - Integrated SNGAN to synthesize visual representations from EEG data, improving interpretability for Brain-Computer Interface (BCI) research.
  - Pipeline applicable to neurofeedback systems and clinical diagnostics.
- **Crater Detection using DETR** | *PyTorch, DETR, DeepMoon Dataset, COCO Format, Roboflow* GitHub
  - Trained DETR on the DeepMoon dataset (Kaguya Lunar imagery) to detect craters using Digital Elevation Maps (DEMs).
  - Preprocessed HDF5 data into COCO format; evaluated with mAP and IoU metrics and visualized attention maps.
  - Enhanced cross-domain generalization by retraining on Roboflow's DOM dataset (Mars and Lunar imagery).
- **Group Success Prediction in EBSNs** | *PyTorch, LSTM, Seq2Seq Models* GitHub
  - Designed an LSTM-based Seq2Seq model to forecast Meetup group success trajectories in event-based social networks.
  - Achieved **87% prediction accuracy**, outperforming baseline models by 10–15%.
  - Insights applicable to community management, social analytics, and event recommendation.

Research Interests

AI, Machine Learning, Computer Vision, NLP, Graph Neural Networks, Robotics, Edge Computing, Depth Sensing, Federated Learning, Distributed Systems, Real-Time Data Processing.

Skills

Languages: Python, C++, JavaScript, SQL, HTML, CSS, Bash

ML & DL: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, LangChain, NLTK

Computer Vision: OpenCV, YOLOv5, Intel RealSense SDK, Open3D, SAM, MediaPipe, DETR

ML Techniques: Object Detection, Segmentation, Image Classification, GANs, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms, RAG, GNNs, Language Modeling, Machine Translation

Web Development: React.js, Node.js, Express.js, Flask, FastAPI, Bootstrap, PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git, Linux, Docker, VS Code, Jupyter, Google Colab, AWS, Hadoop, Spark, Jetson Nano, Raspberry Pi, Arduino

Soft Skills: Leadership, Teamwork, Communication, Problem Solving, Analytical Thinking, Time Management

Awards and Achievements

Hyundai Hope Scholarship *Hyundai Motor India Foundation*

Selected among competitive applicants nationwide for academic excellence, leadership potential, and research contributions. Received **Rs. 1,00,000** in funding to execute an AI-driven project optimizing installation processes, resulting in significant efficiency gains and cost reduction.

Positions of Responsibility

- **Core Member**, DSAI Club, IIT Bhilai — Contributed to research discussions, organized knowledge-sharing sessions, and collaborated on AI-focused initiatives.
- **Core Member**, Ingenuity Coding Club, IIT Bhilai.
- **Member**, Placement Team, IIT Bhilai — Coordinated campus placement drives, streamlined recruiter communication, and ensured smooth execution of interviews.
- **Member**, Event Management / Sponsorship & Outreach Team, IIT Bhilai — Managed logistics, scheduling, and coordination for campus events.
- **Member**, Management Committee, IIT Bhilai.